

Spring Boot with H2 Database



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Spring Boot
2



In Memory
Database

Learn to configure H2 database with [Spring boot](#). H2 database is an in-memory database and is generally used for [unit testing](#) or POC purposes.

Remember that an in-memory database is created/initialized when an application starts up; and destroyed when the application shuts down.

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1. What is H2 Database?

H2 is one of the popular in-memory databases written in Java. H2 can be embedded in Java applications or run in the client-server mode.

Spring Boot provides excellent integration support for H2 using [simple properties configuration](#).

To make itself even more helpful, H2 also provides a console view to maintain and interact with the database tables and data using the GUI interface.

2. Maven Dependency

To use H2 in the Spring boot application, we need to add the H2 runtime jar into dependencies. The best way to add is through maven.

```
pom.xml

<dependency>
  <groupId>com.h2database</groupId>
  <artifactId>h2</artifactId>
  <scope>runtime</scope>
</dependency>
```

3. H2 Configuration Options

3.1. In-Memory Database Configuration

Spring provides effortless configuration options to connect to any database using properties. Below are the configuration properties, we shall have in application.properties file for the most straightforward H2 configuration.

```
application.properties

spring.datasource.url=jdbc:h2:mem:testdb
spring.datasource.driverClassName=org.h2.Driver
spring.datasource.username=sa
spring.datasource.password=
spring.jpa.database-platform=org.hibernate.dialect.H2Dialect
```

Please note by default, spring boot configures the in-memory database connection with the username 'sa' and an empty password. If you wish to change the username and password, override them in the above properties options.

3.2. Persistent Database Configuration

By default, the in-memory databases are volatile, and all stored data will be lost when we restart the application. In this case, data is written in temporary memory, and as soon as JVM is stopped, data is flushed.

To have a persistent data store capable of storing data between application start/stop, we should hold the data in files. For this change the `spring.datasource.url` property.

```
application.properties

# temporary data storage
spring.datasource.url = jdbc:h2:mem:testdb

# temporary data storage
spring.datasource.url = jdbc:h2:file:/data/sample
spring.datasource.url = jdbc:h2:file:C:/data/sample
```

Read More

[H2 database connection URLs](#)

4. Creating Schema and Inserting Data on Initialization

We may want to initialize the database with some fixed schema (DDL) and insert default data (DML) into tables before the application is ready for business usecases.

We can achieve this by putting SQL files into the resources folder (/src/main/resources/).

- **schema.sql** – To initialize the schema ie.create tables and dependencies.
- **data.sql** – To insert default data rows.

```
schema.sql
```

```
DROP TABLE IF EXISTS TBL_EMPLOYEES;  
  
CREATE TABLE TBL_EMPLOYEES (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    first_name VARCHAR(250) NOT NULL,  
    last_name VARCHAR(250) NOT NULL,  
    email VARCHAR(250) DEFAULT NULL  
);
```

```
data.sql
```

```
INSERT INTO TBL_EMPLOYEES (first_name, last_name, email) VALUES  
('Lokesh', 'Gupta', 'abc@gmail.com'),  
('Deja', 'Vu', 'xyz@email.com'),  
('Caption', 'America', 'cap@marvel.com');
```

5. H2 Console

5.1. Enabling H2 Console

By default, the console view of the H2 database is disabled. We must enable it to view and access it in the browser. Note that we can customize the URL of the H2 console, which, by default, is '/h2'.

```
application.properties
```

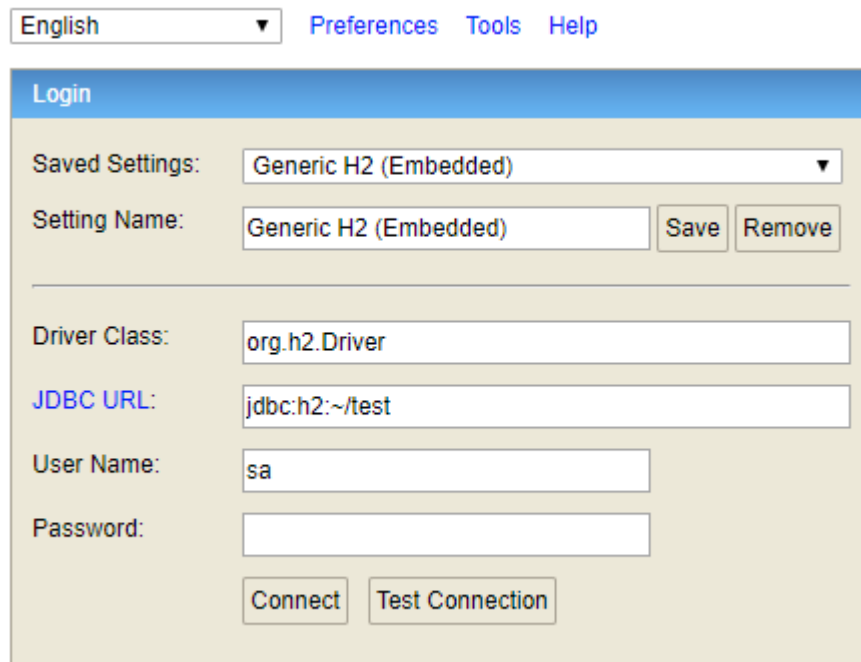
```
# Enabling H2 Console
spring.h2.console.enabled=true

# Custom H2 Console URL
spring.h2.console.path=/h2
```

5.2. Accessing H2 Console

Start the spring boot application and access the console in the browser with this URL : `http://localhost:8080/h2`.

We can see the console like this.



H2 Database Console Login Window

Now enter the configured username and password. We can verify the table structure and default data inserted through SQL files.

The screenshot shows the H2 Console web interface. At the top, there's a toolbar with icons for connection, execution, and settings. Below the toolbar, on the left, is a tree view of the database schema. The main area on the right shows the SQL statement 'SELECT * FROM TBL_EMPLOYEES;' and its result. The result is a table with 4 columns: ID, FIRST_NAME, LAST_NAME, and EMAIL. It contains 3 rows of data. Below the table, it says '(3 rows, 3 ms)' and there is an 'Edit' button.

jdbc:h2:~/test

Max rows: 1000

Auto commit: ☒ Auto complete: Off Auto select: On

Run Run Selected Auto complete Clear SQL statement:

SELECT * FROM TBL_EMPLOYEES;

ID	FIRST_NAME	LAST_NAME	EMAIL
1	Lokesh	Gupta	abc@gmail.com
2	Deja	Vu	xyz@email.com
3	Caption	America	cap@marvel.com

(3 rows, 3 ms)

Edit

H2 Console View

5.3. Other Configuration Options

Spring boot provides two more properties to customize the behavior of the H2 console further. We can enable/disable the database trace logs, and we can **enable/disable the remote access of H2 console**.

By default, both properties are false.

```
application.properties
```

```
# Whether to enable trace output.
```

```
spring.h2.console.settings.trace=false
```

```
# Whether to enable remote access.
```

```
spring.h2.console.settings.web-allow-others=false
```

6. Conclusion

In this **Spring boot with H2 database tutorial**, we learned to configure, initialize and access the H2 database through a spring boot application using simple properties configuration options.

Happy Learning !!

References:

[H2 DB Tutorials](#)

Sourcecode download
